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Lab 2: Intro to Isoline Maps and Earth-Sun Relations

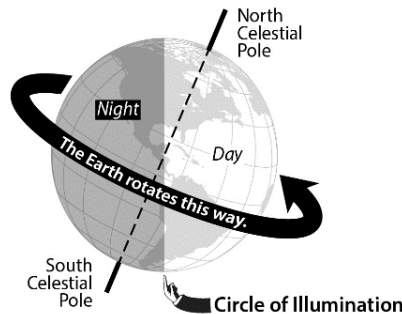
GEOG 5, Instructor: Joy Fritschle

This lab is comprised of 3 parts. In the first part of the lab, you will examine how the Earth's movements in relation to the sun result in the seasons. In the second part of the lab, you will be introduced to isoline maps, an important type of map for studying the four environmental spheres (atmosphere, hydrosphere, biosphere, lithosphere). In the third part of the lab, you'll have the opportunity to look up an online topographic map to enhance your map reading skills.

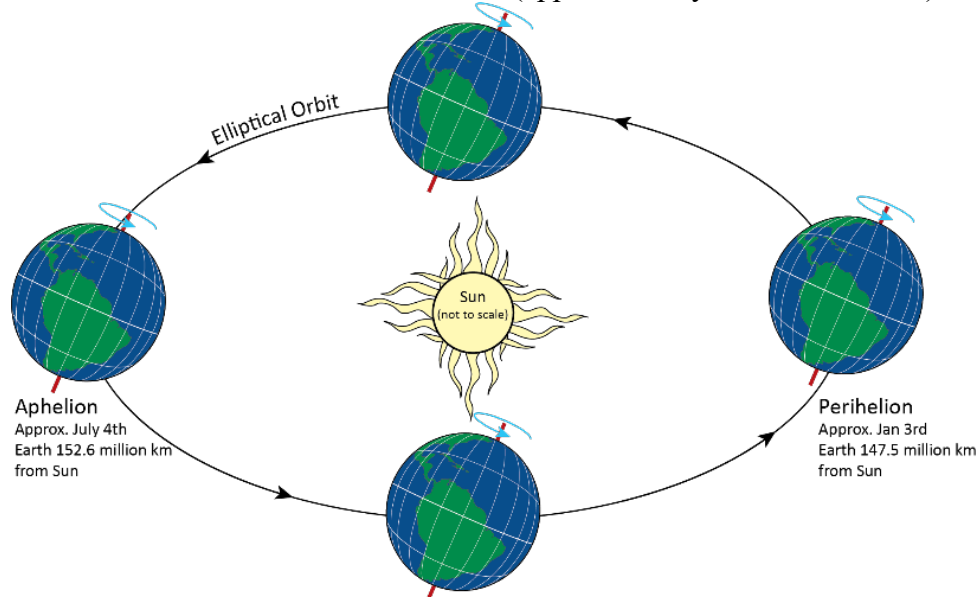
Part 1: Earth-Sun Relations

Earth has five key characteristics that play a role in the Annual March of the Seasons: rotation, revolution, tilt, polarity, and shape.

Rotation: Earth rotates on its axis every 24 hours, which we consider to be one day. Each rotation can be seen in the daily change from day to night. The circle of illumination is the line separating the part of the planet receiving sunlight and the part of the planet in darkness.

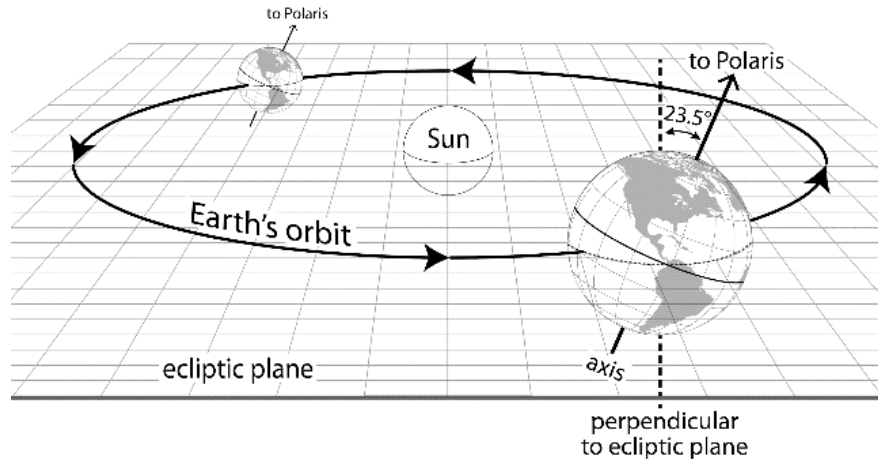


Revolution: Earth revolves around the Sun every 365.25 days, which we consider to be one year. This orbit is not a perfect circle as we might imagine; it is actually an elliptical orbit. Because Earth is traveling in an elliptical orbit, it is closer to the Sun on or around January 3 (known as **perihelion**) than it is on or around July 4 (known as **aphelion**). At perihelion, Earth is 147.5 million kilometers (approximately 91 million miles) from the Sun and at aphelion Earth is 152.6 million kilometers (approximately 95 million miles) from the Sun.

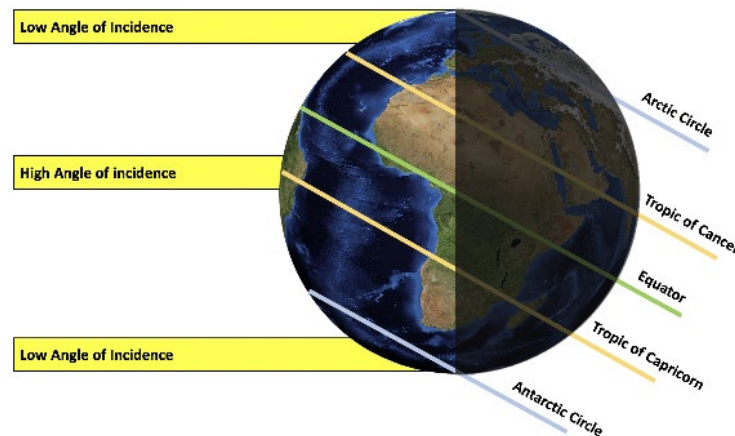


Tilt: Currently, the axis of the Earth is tilted at 23.5° . Earth revolves around the Sun on an imaginary straight line known as the plane of the ecliptic.

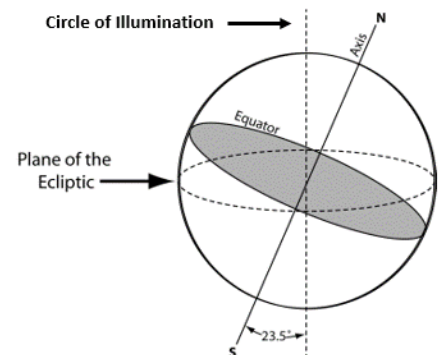
Polarity: While Earth's axis is currently pointing toward Polaris, known as the North Star, the top of the Earth or the North Pole is not always orientated toward the Sun. Sometimes the North Pole is orientated toward the Sun and other times it is orientated away from the Sun. This phenomenon influences the amount of daylight received by the Earth at various latitudes, known as daylength.



Shape: Earth is an oblate sphere (aka **oblate ellipsoid**) and like all spheres, its surface is curved. This means that the Sun's rays strike the Earth at different angles for each latitude. As you can see in the figure below, the Sun's rays strike the Earth at the center (equator) directly, almost at 90° , while they strike toward the poles at a lower angle, more like 10° or 20° . Because the Earth is curved, the **angle of incidence** (the angle of the Sun's rays) varies by latitude. And, because the Earth is tilted, the angle of incidence also varies by season.



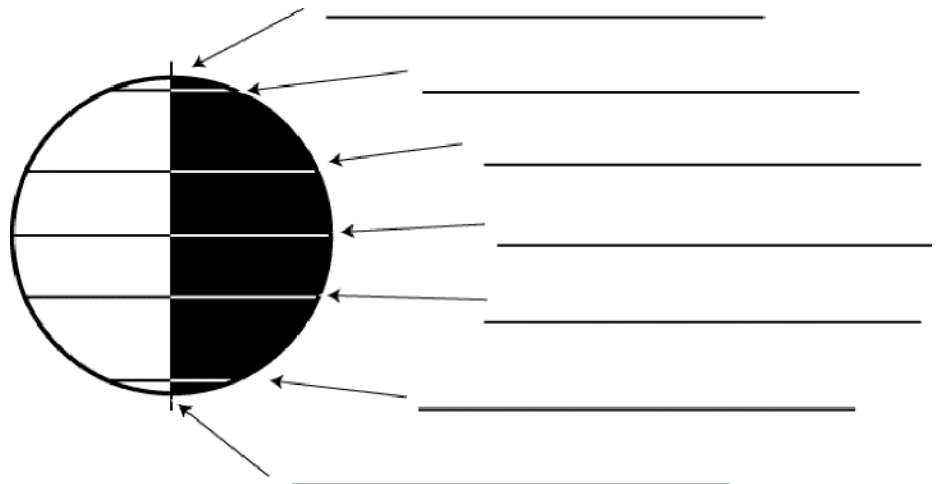
The Sun's vertical rays strike the Earth at the equator (0°) on the **March Equinox** (March 20) and the **September Equinox** (September 21). The March Equinox is known as spring in the northern hemisphere and fall in the southern hemisphere, while the September Equinox is known as fall in the northern hemisphere and spring in the southern hemisphere. During the equinoxes, the **circle of illumination** bisects all parallels (remember that parallels are lines measured in latitude). This even division creates an equal amount of daylight of exactly 12 hours for all latitudes.



The Sun's vertical rays strike the Earth at the Tropic of Cancer (23.5°N), located in the northern hemisphere, on the June solstice (June 22). The Sun's vertical rays strike the Earth at the Tropic of Capricorn (23.5°S), located in the southern hemisphere, on the **December Solstice** (December 21). The **June Solstice** is known as summer in the northern hemisphere and winter in the southern hemisphere, while the December solstice is known as winter in the northern hemisphere and summer in the southern hemisphere. In addition, both hemispheres experience longer daylength in the summer and shorter daylength in the winter. During the December Solstice the Arctic Circle receives 24 hours of darkness, while the Antarctic Circle receives 24 hours of daylight.

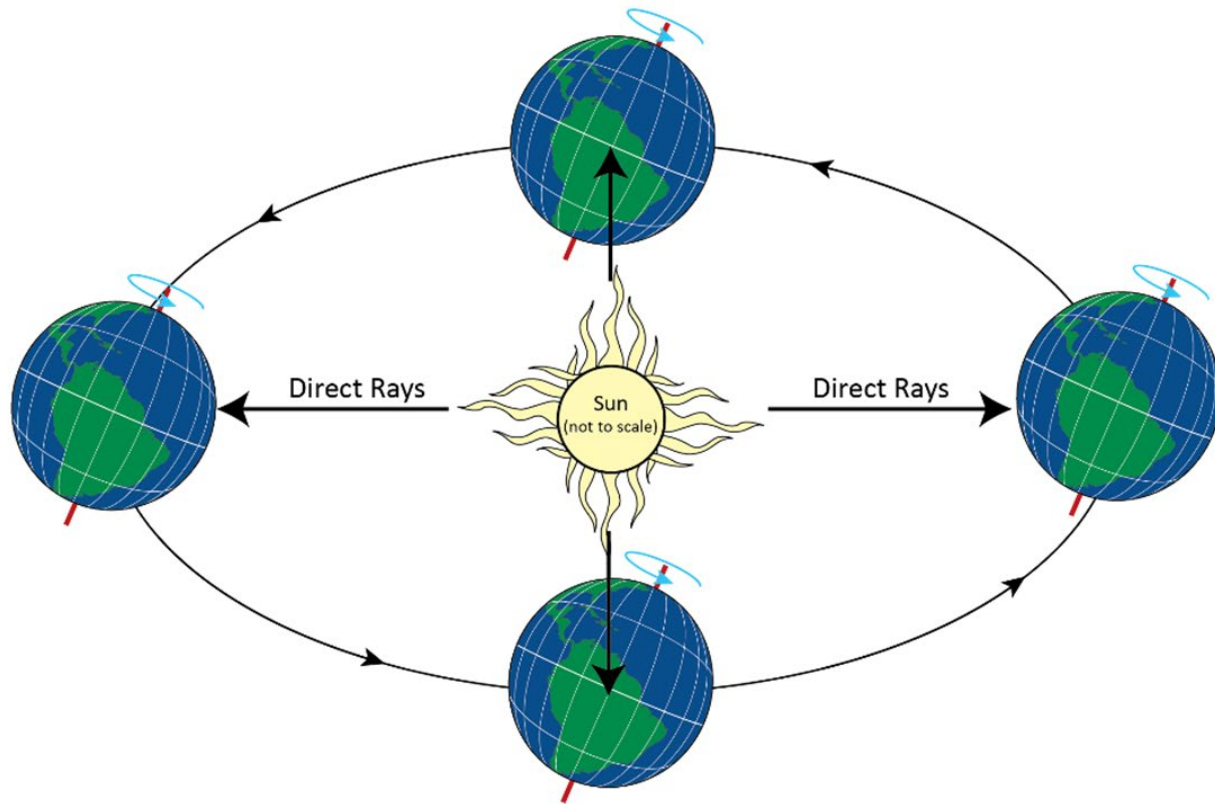
Use what you have learned from the assigned materials (lecture, textbook, and this lab worksheet), and your prior knowledge to answer the questions below.

1. In which hemisphere do you currently live? (1pt)
2. On August 16th: (0.75pts)
 - a. What season are you experiencing?
 - b. What is the daylength like during this season?
 - c. What activities might you do during this season that you might not do in other seasons? Why?
3. On January 16th: (0.75pts)
 - a. What season are you experiencing?
 - b. What is the daylength like during this season?
 - c. What activities might you do during this season that you might not do in other seasons? Why?
4. Identify the seven special parallels of Earth and their latitude. Write each special parallel along with its degree latitude on the correct line on the diagram of the Earth: (3.5pts)
 - Equator, 0°
 - North Pole, 90°N
 - South Pole, 90°S
 - Tropic of Cancer, 23.5°N
 - Tropic of Capricorn, 23.5°S
 - Arctic Circle, 66.5°N
 - Antarctic Circle, 66.5°S



Label the following on the figure below:

5. March Equinox, June Solstice, September Equinox, December Solstice (2pts)
6. Season in northern hemisphere: winter, spring (vernal), summer, fall (autumnal) (1pt)
7. The approximate point of the Earth's revolution in which the Perihelion and Aphelion occur (2pts)



8. Does distance between the Earth and the Sun determine the seasons of the year? Explain. (1pt)

As you have learned, Earth's surface is curved. Therefore, the Sun's rays strike Earth at different angles depending on latitude. Rays that strike Earth directly at a 90° angle are known as **vertical rays**, rays that strike Earth at an angle less than 90° are known as **oblique rays**, and rays that strike Earth at exactly 0° are known as **tangent rays**. Note that there is only one location that experiences vertical rays on a given day, while there are multiple latitudes that experience oblique and tangent rays on a given day.

The latitudes on Earth that receive vertical rays will receive the most intense insolation compared to other latitudes. The latitude of the Sun's vertical rays is called the **declination of the Sun**.

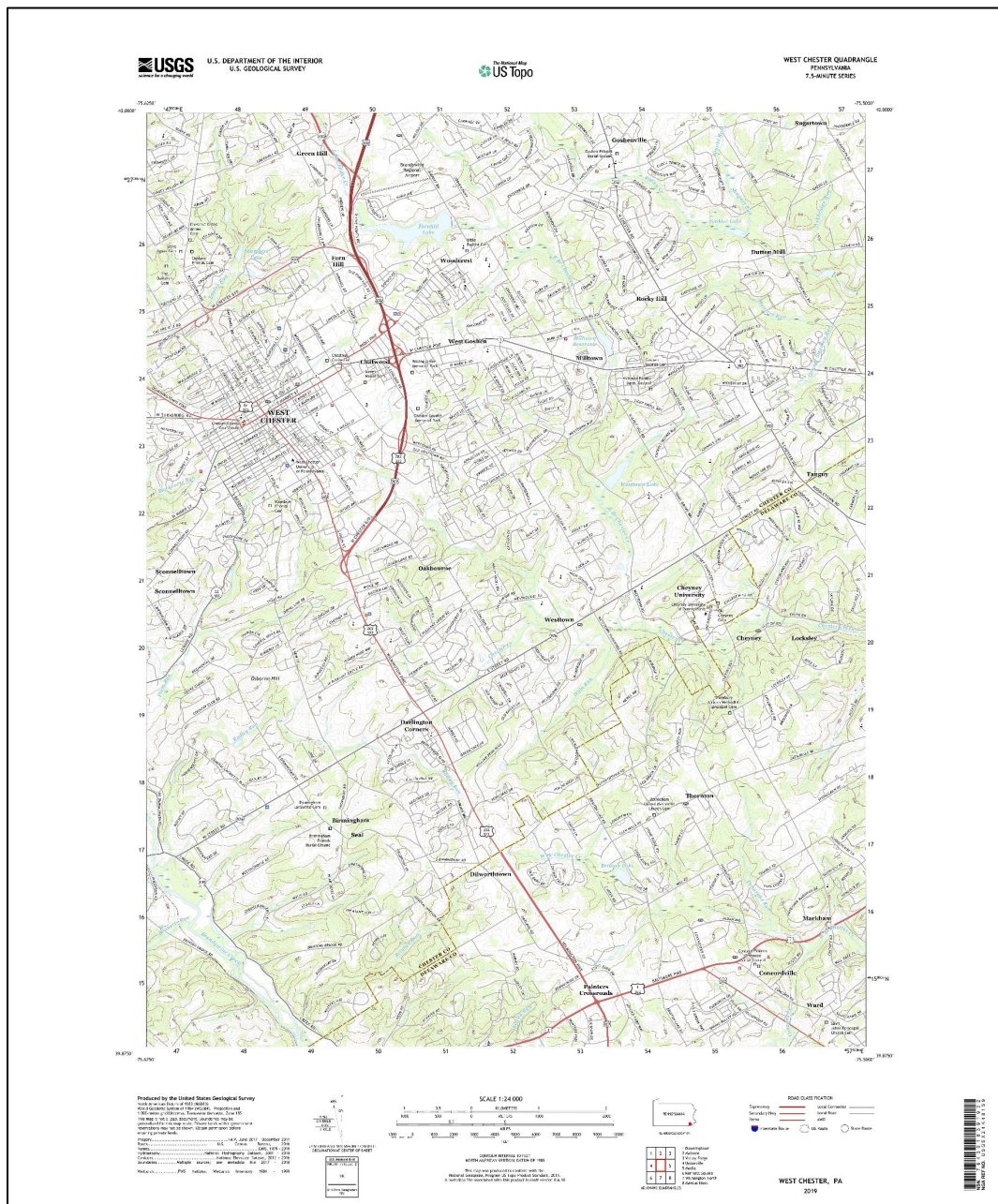
9. What is the declination of the Sun on the March Equinox? On the September Equinox? (1pt)

10. What is the declination of the Sun on the June Solstice? On the December Solstice? (1pt)

11. Do the sun's vertical rays ever reach Santa Monica? (hint: Santa Monica's latitude is 34°N) Why or why not? (2pts)

Part 2: Intro to Isoline Maps

It is very important to be able to learn to interpret topographic maps because it is not always possible to see the area being studied in person. **Topographic maps** depict differences in elevation across landscapes, along with the location of different features such as bodies of water and human settlements (see example below). In order to effectively read a topographic map, you need to understand **map scale** and **isolines**.

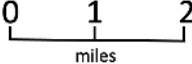


When we visually represent a region of the world on a map, we must reduce its size to fit within the boundaries of the map. **Map scale** measures how much the features of the world are reduced to fit on a map; or more precisely, map scale shows the proportion of a given distance on a map to the corresponding distance on the ground in the real world. The map scale can be represented by a representative fraction (fractional), a verbal scale, or a graphic scale.

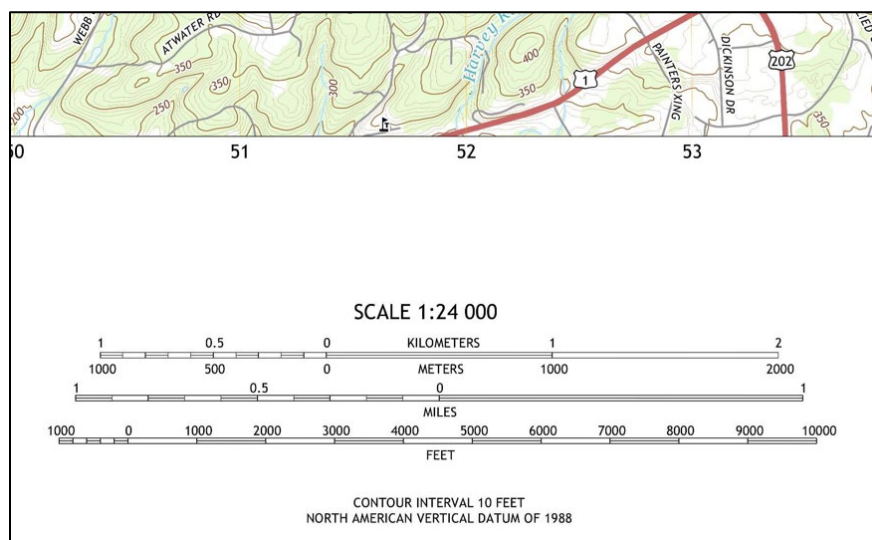
Fractional or Representative Fraction Scale: The most commonly used measure of map scale is the representative fraction (RF), where a map scale is shown as a ratio. With the numerator always set to 1, the denominator represents how much greater the distance is in the world. For example, a map scale with an RF of 1:24000 means that one unit on the map represents 24,000 units on the ground. The representative fraction is accurate regardless of which units are used; the RF can be measured as 1 centimeter to 24,000 centimeters, one inch to 24,000 inches, or any other unit.

Verbal Scale: Some maps use a verbal description of scale, for example, “one centimeter represents one kilometer” or something similar.

Graphic Scale: Scale bars are graphical representations of distance on a map. One important advantage of graphic scales is that they remain true when maps are shrunk or magnified.

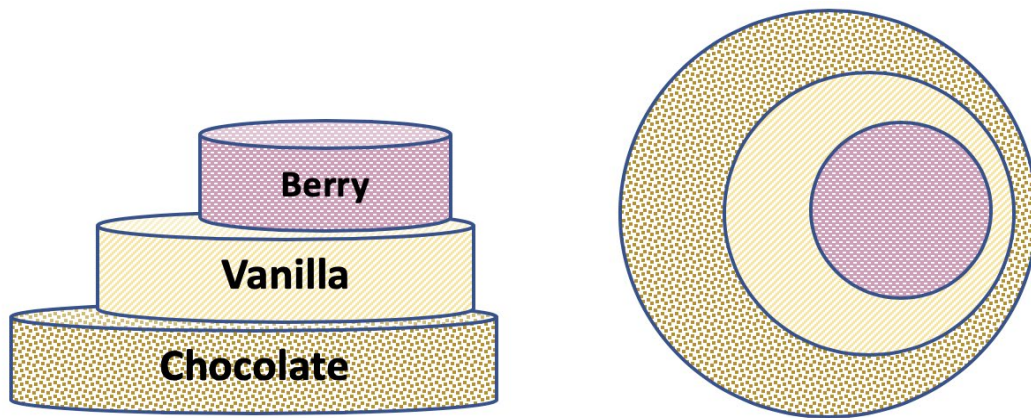
Fractional	1 : 63360
Verbal	1 inch = 1 mile
Graphic	

12. Below is a zoomed in view of the topographic map shown on the previous page. Describe one of these representations of map scale that you see on the map below. If it's a representative fraction or verbal scale, state the scale listed on the map. If it's a graphic scale, describe its appearance. (2pts)



Contour Lines: When observing a topographic map, the first thing observed is the contour lines. Contour lines are continuous lines on a map that join points of equal elevation, meaning that the elevation is constant along the line. Another important element to know is that contours will never cross, or overlap, as they must always remain the same value. Contour lines are especially helpful when interpreting the topography, or the physical shape of the landforms on the map. The spacing between the lines provide the map reader with hints about the slope of the surface.

Here's an easy way to understand how to interpret contour lines: take an object like a three-layered cake. The thickness of each layer is the same, creating its own elevation boundaries or contour lines.



cross-section of the cake seen from the side

a map of the cake showing a bird's-eye view from above

13. Refer to the “cake” figure above on the left, what do you notice about each of the three layers? Do they cross or mix, or are they parallel to one another? Is the cake symmetrical or is it asymmetrical? (1pt)
14. Assume that each layer of cake has a height of 10” (ten inches) and that the table (not pictured) on which the cake is sitting is sea level or 0” (zero inches). Label each isoline shown on the map of the cake (pictured on the right). (2pts)
15. What is the total relief of the cake? (Hint: how tall is the cake?) _____ (1pt)
16. Now imagine that this cake is a hill. Which side of the hill has a less steep gradient? How do you know? (Hint: observe the contour lines or the perimeter of the circles). (2pts)

Topographic maps show lines for selected elevations, otherwise the map would be too confusing to read. The elevation difference between contour lines is called the **contour interval**. If a topographic map has a 10-foot contour interval, that means that each line represents a 10-foot elevation difference between them. Whether the contour interval is a 5-foot, 10-foot, or 100-foot interval depends on the local topography.

17. Refer to the contour lines that you labeled on the map of the cake. What is the contour interval of this map?
(1pt)

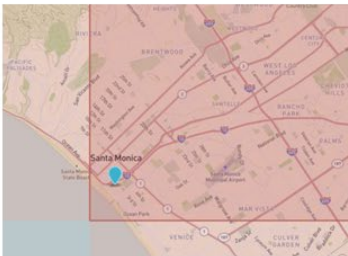
Part 3: Understanding Topographic Maps

To do this part of the lab, you will first need to find a USGS topographic quadrangle to use to answer the questions below. This can either be one of the quadrangles available in class or from the [USGS TopoView website](https://topoview.usgs.gov/)

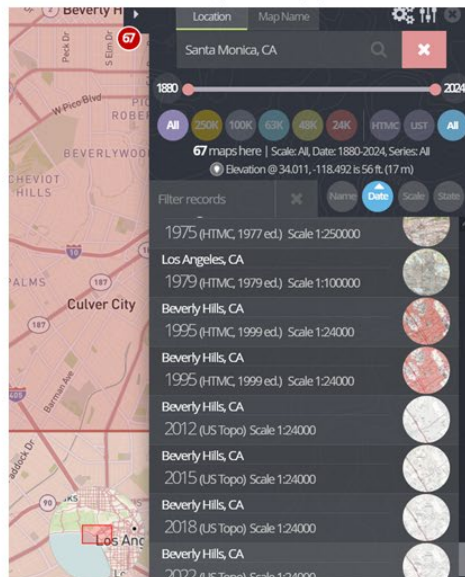
If you choose an online map from USGS Topo View, first select “View and Download maps now,” enter a location in TopoView (pick one of your favorite cities or locations in the U.S.), and select a map on the right (e.g., 2019). Any location and year in the U.S. is fine for this exercise.

Do make sure you're analyzing the actual topographic map, not the map finder window:

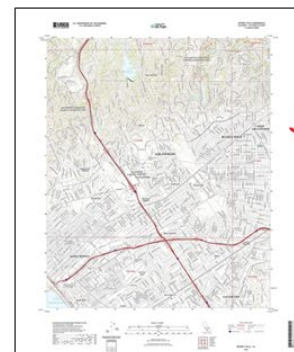
NO; this is the map finder window.



The actual topo maps for various years and places look like this. Choose one to open; any year is fine.



YES; click on a map from a year, then open preview of it or download it (warning: very large files to download so it's better to just preview it).



Study the map and then answer the questions below.

18. What is the name of the map you are using (in other words, what location is this map depicting)? (1pt)

19. What kind of natural features do you see on this map? (2pts)

20. What kind of human features do you see on this map? (2pts)
21. What are the latitude and longitude coordinates for this location? You may do a Google search to help you. Make sure to format your coordinates correctly in degrees, minutes, and seconds. (1pt)
22. What is the scale of your map? You can answer this by either describing the bar scale or the fractional scale. (1pt)
23. This map has contour lines that indicate elevation. Examine these lines on the map, and then try to figure out where on the map you find the highest elevation. What is the highest elevation found on this map and where is it found? (2pts)
24. Lastly, go to timeanddate.com and look up this location. What time is sunrise and sunset at this location today? (1pt)